

Bubble particle heterocoagulation under turbulent conditions

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Abstract

An analytical model that enables the calculation of the flotation rate constant of particles as a function of particle size with, as input parameters, measurable particle, bubble, and hydrodynamic quantities has been derived. This model includes the frequency of collisions between particles and bubbles as well as their efficiencies of collision, attachment, and stability. The generalized Sutherland equation collision model and the modified Dobby–Finch attachment model developed previously for potential flow conditions were used to calculate the efficiencies of particle–bubble collision and attachment, respectively. The bubble–particle stability efficiency model includes the various forces acting between the bubble and the attached particle, and we demonstrate that it depends mainly on the relative magnitude of particle contact angle and turbulent dissipation energy. The flotation rate constants calculated with these models produced the characteristic shape of the flotation rate constant versus particle size curve, with a maximum appearing at intermediate particle size. The low flotation rate constants of fine and coarse particles result from their low efficiency of collision and low efficiencies of attachment and stability with gas bubbles, respectively. The flotation rate constants calculated with these models were compared with the experimental flotation rate constants of methylated quartz particles with diameters between 8 and 80 μm interacting with gas bubbles under turbulent conditions in a Rushton flotation cell. Agreement between theory and experiment is satisfactory.

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1. Introduction

Bubble–particle capture is of particular importance to froth flotation. The dynamics of adsorption at liquid interfaces also influence bubble–particle capture [1]. The stability of the aggregate is essentially one of thermodynamics, whereas dynamic processes control the probability that a bubble and particle will form an aggregate. Let us reflect on these processes briefly.

For capture to occur between a bubble and a hydrophobic particle, they must first undergo a sufficiently close encounter, a process that is controlled by the hydrodynamics governing their approach in the aqueous environment in which they are normally immersed. Should they approach quite closely, within the range of attractive surface forces, the intervening liquid film between the bubble and particle will drain, leading to a critical thickness at which rupture occurs. Movement of the three-phase contact line (the

boundary between the solid particle surface, receding liquid phase, and advancing gas phase) then occurs, until a stable wetting perimeter is established. This sequence of drainage, rupture, and contact line movement constitutes the attachment process. A stable particle–bubble union is thus formed. The particle may only be dislodged from this state if it is supplied with sufficient kinetic energy to equal or exceed the attachment energy; thus a process of detachment can occur.

The collection (or capture) efficiency E_{coll} of a bubble and a particle may be defined as

$$E_{\text{coll}} = E_c E_a E_s, \quad (1)$$

where E_c is the collision efficiency, E_a is the attachment efficiency, and E_s is the stability efficiency of the bubble–particle aggregate. This dissection of capture efficiency into three processes, proposed many years ago, focuses attention on the three regions of bubble–particle capture around the bubbles where, in order, hydrodynamic interactions, surface forces, and forces controlling bubble–particle aggregate stability are dominant [2]. It is now possible to describe these various processes with at least acceptable reliability and to predict the efficiency of bubble–particle capture for single bubbles as they rise through a dilute pulp under nonturbulent conditions [3–5]. However, a major challenge is to describe

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the process of bubble–particle capture as a bubble swarm passes through a concentrated particle dispersion when turbulence is present. In such circumstances, numerous bubble–particle encounters occur and the task is to take the specific processes that occur between individual bubbles and particles and merge them with the multiple bubble–particle encounters occurring within a turbulent environment. This is the focus of this present paper, where we develop the background theory, identify general trends, and test the theory against experimental data obtained for a model system of quartz particles. The application of this approach to mineral and ore systems is dealt with elsewhere [6].

2. Theory

The basis for our understanding today of the process of particle collisions in fluids is a result of the efforts of Smoluchowski [7], who considered particle aggregation to be equivalent to a series of chemical reactions; he developed equations describing particle aggregation rates as well as expressions for the rate of particle contacts in solution. The coagulation of colloidal particles is considered to be a two-step process. The first is the physical collision of two or more particles or aggregate, while the second process is one of attachment. In flotation, detachment constitutes the third process.

The process of particle collision is governed by physical factors such as diffusion, temperature, fluid shear, particle and fluid density, and the sizes of particles and aggregates. Whether particles will adhere when they collide is a function of conditions at the interface between the particles and the fluid medium. Chemical interactions at the solid–liquid interface (and air–liquid interface) are responsible for the development of the surface charge and potential, the electric diffuse layer, and hydration and hydrophobic effects which determine the probability of particle attachment. For bubble–particle encounters, detachment is determined by the net adhesive force, i.e., the sum of the attachment forces (e.g., capillary) minus the detachment forces (e.g., acceleration) [8].

For a suspension of particles in a fluid moving under uniform shear (laminar flow), if the particles are randomly distributed throughout the fluid, following the fluid streamlines up to the moment of impact, their collision frequency, Z_{ij} (the number of collisions per unit volume per unit time between particles of type i and j , with number densities N_i and N_j), can be represented by

$$Z_{ij} = \frac{4}{3} N_i N_j d_{ij}^3 G, \quad (2)$$

where d_{ij} is the sum of the particle and bubble radii and G is the velocity gradient in the fluid perpendicular to the direction of particle movement. We now see the impact of the fluid on the collision frequency, providing us with an inkling as to where turbulence might be introduced. Camp and Stein [9] were the first to apply Eq. (2) to industrial

processes, dealing with the case of flocculation. They used the mean velocity gradient in a turbulent fluid, $\bar{G}$, instead of G . $\bar{G}$ was related to the energy dissipated per unit mass of fluid, or Kolmogorov energy density, ε , and the kinematic viscosity, ν , by

$$\bar{G} = \left(\frac{\varepsilon}{\nu} \right)^{1/2}. \quad (3)$$

Thus

$$Z_{ij} = \frac{4}{3} N_i N_j d_{ij}^3 \left(\frac{\varepsilon}{\nu} \right)^{1/2}. \quad (4)$$

Saffman and Turner [10] obtained a result similar to that of Camp and Stein, although the constant in Eq. (4) was slightly smaller ($\sim 3\%$). This mechanism of collision under a shear gradient is valid when d_{ij} is small relative to the smallest eddies in the fluid and for particles which follow the fluid motion. Turbulence with low energy dissipation is assumed. The process is confined within the smallest eddies and assumes that the particle paths relative to an eddy are determined completely by the eddy fluid velocities and accelerations.

Equation (4) is not valid, then, for large particles and/or more vigorous fluid turbulence, typically found in industry. In these circumstances, the fluid and particle velocities do not coincide. In larger scale industrial situations involving commercial flotation separations, particles acquire momentum in one or more eddies and then will be projected into a neighboring eddy.

Based on pioneering work by Batchelor [11], as well as Levins and Glastonbury's solution [12] of the Tchen equation for a particle accelerating relative to a fluid, Abrahamson [13] evaluated the root mean square (rms) velocities of particles relative to the mean fluid velocity, $(\bar{V}_p^2)^{1/2}$, in a suspending fluid where the turbulence was isotropic on the scale of the collision process and where the particles were projected at one another from independently moving elements of fluid (high Reynolds numbers). The collision frequency can then be calculated, and for particles colliding with bubbles is given by

$$Z_{pb} = 5 N_p N_b d_{pb}^2 \sqrt{\bar{V}_p^2 + \bar{V}_b^2}, \quad (5)$$

where N_p and N_b are the number densities of particles and bubbles, respectively, $d_{pb} = (d_p + d_b)/2$, and $\bar{V}_p$ and $\bar{V}_b$ are the velocities of the particles and bubbles relative to the fluid, respectively.

The rms relative velocities of the particles and bubbles may be calculated from the specific power input using an equation generated originally by Liepe and Mockel [14], shown here for a particle, and used by others for the description of turbulent processes in flotation cells [15,16],

$$(\bar{V}_p^2)^{1/2} = \frac{0.33 \varepsilon^{4/9} d_p^{7/9}}{\nu^{1/3}} \left(\frac{\Delta \rho_p}{\rho_{fl}} \right)^{2/3}, \quad (6)$$

where $\Delta \rho_p = \rho_p - \rho_{fl}$, the difference between the particle and fluid densities.

There are restrictions on the upper and lower limits of particle size for using Eqs. (4)–(6). In water, one of the colliding “particles” must have a diameter greater than 100 μm (e.g., the bubble) [13]. Other approximations involve Reynolds number ranges (assumed high) and the exact nature of the movement of bubble–particle aggregates in eddies (these aggregates are taken to always move outward in an eddy but if their density is less than that of the fluid the reverse may occur [17]). Hence these equations are necessarily approximate.

If $d_b \gg d_p$, then d_p and $\bar{V}_p^2$ can be neglected and $d_{pb} \sim d_b/2$. Thus

$$Z_{pb} \approx 5N_p N_b \left(\frac{d_b}{2} \right)^2 \left[\frac{0.33\epsilon^{4/9} d_b^{7/9}}{\nu^{1/3}} \left(\frac{\Delta\rho_b}{\rho_{fl}} \right)^{2/3} \right]. \quad (7)$$

We learn that the number of bubble–particle collisions per unit volume per unit time, Z_{pb} , is equal to the product of the number densities of particles and bubbles at any time, $N_p N_b$, multiplied by terms containing frequency, bubble size, ϵ , ν , and ρ . Hence the influence of turbulence is amply demonstrated. We now see how the expression for “flotation rate” may be modified accordingly.

We commence (e.g., [18,19]) with the rate of removal of particles by bubbles,

$$\frac{dN_p}{dt} = -zN_p N_b E_{coll}, \quad (8)$$

where E_{coll} is given by Eq. (1) and z is the frequency.

Recall that

$$Z_{pb} = zN_p N_b \quad (9)$$

and

$$\frac{dN_p}{dt} = -kN_p, \quad (10)$$

where

$$k = zN_b E_{coll}. \quad (11)$$

The dimensions of k are time^{-1} . N_p and N_b are the numbers of particles (N) and bubbles per unit volume, V , respectively ($N_p = N/V$). N_b can be related to the gas flow rate, G_{fr} , and the residence time, t_r , of bubbles in the unit volume,

$$N_b = \frac{6G_{fr}}{\pi d_b^3 V_{cell}} t_r, \quad (12)$$

where V_{cell} is the volume of the flotation cell. The residence time, t_r , can be expressed as the time that the bubbles of velocity v_b remain in the unit volume,

$$t_r = \frac{1 \text{ unit length}}{v_b}, \quad (13)$$

where v_b is the average bubble velocity in the unit volume.

Combining Eq. (8) with Eqs. (7) and (9) we have

$$\frac{dN_p}{dt} = -N_b 5 \left(\frac{d_b}{2} \right)^2 \left[\frac{0.33\epsilon^{4/9} d_b^{7/9}}{\nu^{1/3}} \left(\frac{\Delta\rho_b}{\rho_{fl}} \right)^{2/3} \right] E_{coll} N_p, \quad (14)$$

where the term

$$5 \left(\frac{d_b}{2} \right)^2 \left[\frac{0.33\epsilon^{4/9} d_b^{7/9}}{\nu^{1/3}} \left(\frac{\Delta\rho_b}{\rho_{fl}} \right)^{2/3} \right]$$

in Eq. (14) (equivalent to the term $5d_{pb}^2 \sqrt{\bar{V}_p^2 + \bar{V}_b^2}$ in Eq. (5)) is the collision volume of fluid that is swept by the bubble per unit time.

Substituting for N_b from Eqs. (12) and (13), we obtain

$$\begin{aligned} \frac{dN_p}{dt} = & \overbrace{-2.39 \frac{G_{fr}}{d_b V_{cell}}}^{\text{mechanical term}} \overbrace{\left[\frac{0.33\epsilon^{4/9} d_b^{7/9}}{\nu^{1/3}} \left(\frac{\Delta\rho_b}{\rho_{fl}} \right)^{2/3} \frac{1}{v_b} \right]}^{\text{primary turbulence term}} \\ & \times \underbrace{E_c E_a E_s N_p}_{\text{elementary processes}}. \end{aligned} \quad (15)$$

As a guide, a fairly typical value for the “mechanical term” in Eq. (15) is $46 \times 10^{-2} \text{ m}^{-1} \text{ min}^{-1}$ and $2 \times 10^{-2} \text{ m}$ for the “primary turbulence term” [20].

There is a strong similarity to

$$\frac{dN_p}{dt} = -\frac{3}{2} \frac{G_{fr} h}{d_b V_{cell}} E_c E_a N_p, \quad (16)$$

the expression applying to a batchwise flotation process in the absence of turbulence [18,19]. In this case h is the height of the flotation cell and, E_s is unity. We also note that the expressions for the superficial gas velocity, J_g , and the superficial surface area rate of bubbles, S_b , now used commonly in flotation cell characterization, can be readily introduced into Eq. (15) [21,22].

2.1. Collision efficiency, E_c , dominated by bulk hydrodynamics

It is necessary to consider the basic equations of the fluid flow regime around the surface of a bubble and recognize that, under flotation conditions, the bubble surface will be mobile or very close to this situation [4,23,24]. In the case of a moving bubble, even if adsorbed surfactant was present under static conditions, it is swept to the rear, so that the front surface is mobile [1,24]. The condition of potential flow (high Re_b) applies in this present study.

The key equation that must be solved to describe the deposition of particles from a flow onto a spherical bubble surface is the Basset–Boussinesq–Oseen (BBO) equation. In recent publications we have shown how this is achieved as well as the good agreement that is obtained when individual bubble collection efficiencies, obtained from single bubble experiments, are compared with theoretical predictions [4,5, 25–27]. Our specific solution of the BBO equation is termed the generalized Sutherland equation (or GSE) and takes account of both interception and inertial forces. When inertial forces are introduced, one needs to recognize that hydrodynamic pressing and centrifugal forces act upon the particle. Both of these forces, and hence the resultant *total* force,

are angle dependent. There is, therefore, a maximum possible collision angle, θ_t , called the angle of tangency, beyond which collision is prevented.

In solving the BBO equation, we have

$$\begin{aligned}\cos \theta_t &= (1 + \beta^2)^{1/2} - \beta, \\ \sin^2 \theta_t &= 2\beta[(1 + \beta^2)^{1/2} - \beta].\end{aligned}\quad (17)$$

The dimensionless number β is defined as [4]

$$\beta = \frac{4E_{c-su}}{9K_3}, \quad (18)$$

where E_{c-su} is the Sutherland [28] E_c formula, $\sim 3d_p/d_b$, accounting only for interception and K_3 is defined as

$$K_3 = \frac{2v_b(\rho_p - \rho_{fl})r_p^2}{9\nu r_b}, \quad (19)$$

and where β is a measure of the relative importance of the interceptional and inertial contributions to the collision process. The final solution of the BBO equation is the GSE,

$$\begin{aligned}E_c &= E_{GSE} \\ &= E_{c-su} \sin^2 \theta_t \exp \left\{ \left[3K_3 \left(\ln \frac{3}{E_{c-su}} - 1.8 \right) \right. \right. \\ &\quad \left. \left. - \frac{4(\frac{2}{3} + \frac{\cos^3 \theta_t}{3} - \cos \theta_t)}{\sin^4 \theta_t} \right] \cos \theta_t \right\}.\end{aligned}\quad (20)$$

This GSE may be used to predict the collision efficiencies of bubble–particle collisions with reliability and is applicable under conditions of potential flow over the bubble surface (high Reynolds numbers for the bubble). However, there is one cautionary note at this point. Turbulence may cause specific modes of particle transport toward the bubble surface that are not accounted for in present solutions of the BBO equation [29]. As far as we are aware, the literature is silent both experimentally and theoretically on these issues at present, so that there is a task defined for the future.

2.2. Attachment efficiency, E_a , dominated by interfacial behavior

In single bubble experiments, the experimental attachment efficiency can be determined from the measured collection efficiency, E_{coll} , under conditions where the stability efficiency is unity and E_c is calculated [25,27,30,31]; thus

$$E_a = \frac{E_{coll}}{E_{c1}}. \quad (21)$$

The attachment model assumes that particle–bubble collision occurs evenly over the section of the bubble surface between angles $\theta = 0$ and $\theta = \theta_t$ where the angle θ is measured from the upper surface of the rising bubble, and the angle θ_t is the maximum possible collision angle as shown in Fig. 1.

A modified Dobby–Finch attachment model [32] is used to calculate the attachment efficiency (E_a), defined as the

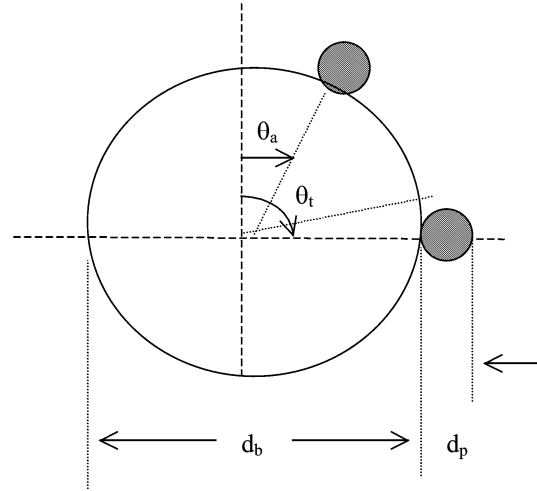


Fig. 1. Schematic representation of a particle of diameter d_p sliding around a bubble of diameter d_b from a collision or adhesion angle, θ_a , to a maximum sliding angle, $\pi/2$. θ_t is the maximum possible collision angle or angle of tangency.

ratio of the projected area corresponding to an angle θ_a , to the projected area corresponding to θ_t . The basic equation of this model is mathematically expressed as

$$E_a = \frac{\sin^2 \theta_a}{\sin^2 \theta_t}, \quad (22)$$

where θ_a , termed the adhesion angle, is the specific collision angle where, if a particle collides at this angle, its sliding time equals the induction time. The angle θ_a relates both the sliding and induction times to the attachment efficiency. The maximum collision angle, θ_t , is a complex function of the bubble Reynolds number and satisfies Eqs. (17)–(19).

Combining Eq. (18) and (19) with the expression $v_b d_b \rho_{fl} / \nu = Re_b$, we can express β as a function of the Reynolds number, Re_b ,

$$\beta = \frac{12d_b \rho_{fl}}{d_p(\rho_p - \rho_{fl}) Re_b}. \quad (23)$$

As mentioned above, for a particle that collides with the bubble surface at a certain collision angle and then slides along the surface toward the equator of the bubble for a sliding time which is just equal to the induction time, this particular collision angle is the adhesion angle (θ_a).

Since the sliding velocity, or sliding time, of a particle depends on, among other factors, the fluid flow regime at the bubble surface, the establishment of the sliding time and attachment efficiency models have to be based on the particular fluid flow condition. Recall that so long as the bubble size is not too small and that the concentration of the surface active materials is not too high, the bubble surface, especially its upper half where particle–bubble collision and attachment occur, is mobile. Under these conditions, a high bubble rising velocity can be achieved and the fluid flow at the bubble surface is essentially that of potential flow.

Sutherland [28] and later Dobby and Finch [32] derived an expression for the sliding time t_{sl} of a particle colliding

with a bubble at a collision angle, θ , to a maximum sliding angle, $\theta = \Pi/2$, under potential flow conditions:

$$t_{sl} = -\frac{d_p + d_b}{2(v_p + v_b) + v_b\left(\frac{d_b}{d_p + d_b}\right)^3} \ln\left(\tan \frac{\theta}{2}\right). \quad (24)$$

This equation can be used to calculate the adhesion angle, θ_a , in Eq. (22), for this adhesion angle is the collision angle where sliding time equals induction time, t_{ind} :

$$\theta_a = 2 \arctan \exp\left[-t_{ind} \frac{2(v_p + v_b) + v_b\left(\frac{d_b}{d_p + d_b}\right)^3}{d_p + d_b}\right]. \quad (25)$$

There is ample experimental and theoretical evidence [33,34] to show that the induction time is related to the particle size by the empirical equation

$$t_{ind} = Ad_p^B. \quad (26)$$

Equation (22) together with Eqs. (17), (23), (25), and (26) is used to calculate E_a as a function of the known parameters d_p , d_b , v_p , v_b , ρ_p , ρ_{fl} and the unknown parameters A and B . The calculated attachment efficiencies versus particle size curves are then compared to the experimental curves of E_a versus d_p , obtained under defined contact angle, ionic strength, and pH conditions, and a nonlinear least-squares program is used to optimize the values of the parameters A and B until the best fit of the experimental curves is obtained [35]. This approach is certainly not perfect at this time. Ideally the calculation of E_a should embrace thin film drainage between a bubble and a heterogeneous solid surface, including possible bubble deformation, film rupture, and the movement of the three-phase contact line over the surface. The induction time, t_{ind} , really consists of three components—film drainage, film rupture, and three-phase contact line spreading—and is inversely proportional to the contact angle of the particles [35]. At this stage, although much is known about these individual events [36,37] it is impossible to describe them all quantitatively.

2.3. Particle–bubble stability efficiency: hydrodynamics and interfacial behavior are both important

The probability of aggregate stability depends on the attachment force between the bubble and the particle in relation to the external stress forces in the environment. If the attachment force is greater than the sum of all stress forces, then this aggregate remains stable on its long way from the pulp phase to the froth phase. The force balance for a spherical particle in the liquid/gas interface can now be described in a physically clear manner [8,38,39],

$$F_{ca} + F_{hyd} + F_b - F_g - F_d - F_\sigma = 0, \quad (27)$$

where the forces acting upon the particle are the capillary, hydrostatic, buoyancy, gravitational, machine acceleration and capillary pressure terms, respectively, the sum of which is zero at equilibrium.

The ratio of the detachment forces F_{det} to the attachment forces F_{att} characterizes the aggregate stability. This ratio is a dimensionless parameter, B_0^* , analogous to the Bond number:

$$B_0^* = F_{det}/F_{att} = \frac{F_g - F_b + F_d + F_\sigma}{F_{ca} + F_{hyd}}. \quad (28)$$

The equation for the Bond number may be derived as

$$B_0^* = \frac{d_p^2(\Delta\rho_p g + \rho_p a) + 1.5d_p(\sin^2 \omega)f(d_b)}{|6\gamma \sin \omega \sin(\omega + \varphi)|} \quad (29)$$

with

$$f(d_b) = \left(\frac{4\gamma}{d_b} - d_b\rho_{fl}g\right), \quad (30)$$

where φ is the contact angle, $\omega = 180^\circ - \varphi/2$, refers to the location of a particle at the liquid–vapor interface and γ is the surface tension [38,40].

The additional acceleration, a , which determines the detachment forces, depends on the structure and the intensity of the turbulent flow field thus on the dissipation energy, ε , in a given volume of apparatus. Schulze [41] assumed that aggregates, the dimensions of which correspond to those of the turbulent vortices in the inertial region, are moved mainly by the centrifugal acceleration $a = a_{cen}$ present in the vortex. If r_v is the radius of such a vortex and $(\bar{V}_v^2)^{1/2}$ is its rms velocity, then a_{cen} in the inertial region is given by

$$a_{cen} \approx \frac{(\bar{V}_v^2)}{r_v} = \frac{(1.38(\varepsilon r_v)^{1/3})^2}{r_v} = 1.9 \frac{\varepsilon^{2/3}}{r_v^{1/3}}. \quad (31)$$

For aggregates, where the particle size is smaller than the bubble size, the vortex radius, r_v , should be set equal to the aggregate radius. Hence,

$$a_{cen} \approx 1.9 \frac{\varepsilon^{2/3}}{\left(\frac{d_b}{2} + \frac{d_p}{2}\right)^{1/3}}. \quad (32)$$

If the bubble–particle aggregate stability efficiency, E_s , is exponentially distributed [8] then E_s can be determined from the equation

$$E_s = 1 - \exp\left(1 - \frac{F_{att}}{F_{det}}\right) \quad (33)$$

and may also be expressed in terms of the “Bond number,” B_0^* , using Eq. (29)

$$E_s = 1 - \exp\left(1 - \frac{1}{B_0^*}\right). \quad (34)$$

In turbulent flow, various eddy sizes can be observed [16,42] and may be divided into three categories. The eddy size may be several times smaller than the bubble–particle aggregate; several times larger than the bubble–particle aggregate; and of an order similar to that of the bubble–particle aggregate.

In the first case, the eddy does not have sufficient energy to influence the bubble–particle aggregate detectably. In the second instance, the size of the eddy is much larger

than that of the bubble–particle aggregate, and the aggregate is captured by such an eddy. During this process the bubble–particle aggregate is moved by the eddy flow under the drag force. The bubble–particle aggregate moves around the eddy center and centrifugal force is produced on the particle, which is the main detachment force between the bubble and its attached particle [13].

The third situation occurs when the bubble–particle aggregate size is similar to that of the eddy. In this case the bubble–particle aggregate is unable to be captured by an eddy. Instead, the elastic bubble acts like an eddy of similar dimensions and a dynamic interaction occurs between the aggregate and these eddies. This interaction causes the bubble to vibrate with the same frequency of energy transfer, and the vibration of the bubble in turn causes the attached particles to circulate on the bubble surface with a frequency equal to the frequency of vibration of the bubble. Detachment forces, described above, are therefore generated between the bubble and its attached particles. In a mechanical, agitated flotation cell, the dissipation energy near the impeller zone is usually 10–100 times higher than the mean dissipation energy. Consequently, for large particles, the bubble–particle aggregates formed would not survive in this high energy dissipation region. Schulze [41] suggested that the *mean vessel energy dissipation is more decisive in determining the mean stability of aggregates, rather than the local dissipation energy*. Hui [43] noted that the average acceleration of the attached particle, $b_{p\max}$, may be expressed as a function of the mean dissipation energy, ε , obtaining

$$b_{p\max} = 29.6 \frac{\varepsilon^{2/3}}{d_{\text{agr}}^{1/3}}, \quad (35)$$

where d_{agr} is the aggregate diameter and where the coefficient in Eq. (35) is larger than that for Eq. (31), derived by Schulze [41].

From Eq. (31) or (35) we deduce that an increase in the power input will result in a decrease in the maximum floatable particle size. This is, of course, consistent with intuition and observations reported in the literature [44].

With appropriate expressions for the calculation of E_c , E_a , and E_s , we thus have a method, using Eq. (15), for calculating the bubble–particle heterocoagulation rate constants under turbulent conditions.

3. Experimental materials and methods

3.1. Reagents and particles

Milli-Q water (Millipore Corp., Bedford, USA) prepared by reverse osmosis of tap water, followed by two stages of ion exchange and two stages of activated carbon adsorption, with a final filtration through a 0.22- μm filter was used. This water has a surface tension of 72.8 mN m⁻¹ at 20 °C and a pH of 5.6 $\pm$ 0.1. All chemicals were of analytical reagent grade quality unless otherwise specified. All the glassware

was washed with nitric acid (50:50) and rinsed thoroughly with Milli-Q water before use.

Trimethylchlorosilane (TMCS) (>99.95%, Merck) and cyclohexane (99.95% HPLC grade) were stored in glass containers and kept under a dry nitrogen environment (99.99%, CIG) until needed for methylation of quartz particles and contact angle determination. Polypropylene glycol (MW250, Aldrich) was used as the frother in all experiments. KOH and HNO₃ were used as pH modifiers whilst KNO₃ was used as an indifferent electrolyte.

3.2. Quartz particles

High-purity optical grade quartz obtained from Tenterfield, NSW, was crushed, wet-ground, and then passed through a –38- μm sieve. The +38- and –38- μm quartz particles were then cleaned with hot aqua regia for 2 h to remove any surface mineral impurities. The acid was decanted and the quartz was then rinsed thoroughly with Milli-Q water. The quartz was then immersed in hot 30% KOH for 30 s to remove organic contamination. The alkaline supernatant was then decanted and the quartz washed again with Milli-Q water until the pH became neutral. The particles were found to be clean using the bubble cling technique [45]. The quartz particles were then dried at 110 °C in a clean oven for 24 h and then stored subsequently in sealed glass containers in a clean vacuum desiccator over fresh silica gel. Quartz particles were methylated to varying degrees of hydrophobicity by immersion in solutions of TMCS in cyclohexane, following methods described in detail previously [39,46]. Advancing water contact angles on these particles were determined using the Washburn and equilibrium capillary pressure techniques [47,48].

3.3. Flotation cell

A Rushton turbine flotation cell of defined geometry and known hydrodynamics [49–51] was constructed. Its standard dimensions include the height, which should be equal to its diameter. The impeller diameter is typically one-third of the diameter of the cell, while the blade height and width are one-fifth and one-quarter the length of the impeller diameter, respectively. Four baffles, equispaced with a width one-tenth the diameter of the cell, aid in reducing the vortex formed at the pulp/froth interface. The Rushton turbine cell generates a flow, leaving the impeller, in both radial and tangential directions. This radial–tangential jet flow divides at the vessel wall, and the flow then recirculates back into the impeller region [51]. The exact specifications are given in the Table 1.

The flotation cell was powered by an overhead variable speed stirrer. This system enabled the impeller speed to be controlled accurately; the impeller speed was confirmed independently using an optical tachometer.

Rushton turbines are high-shear axial-flow impellers with high power numbers and are commonly used for mixing and gas dispersion in industry. Rushton turbines are similar in

Table 1
Rushton turbine flotation cell

Tank specifications	Impeller specifications
Volume = $2.25 \times 10^{-3} \text{ m}^3$	Six-blade Rushton turbine
Diameter = 0.144 m	Diameter = 0.048 m
Height:diameter = 1:1	Width:height:diameter = 5:4:20
Baffle:diameter = 1:10	

both principle and characteristics to the impellers used in industrial mechanical flotation cells, for there is a general trend toward the use of high-shear axial-flow impellers with high power numbers in these cells [52]. The impeller was located at a standard distance of one impeller diameter from the bottom of the tank (one-third of the height of the cell). The tank was aerated by nitrogen gas or compressed air, fed through a stainless steel 13- μm porous frit set into the base of the tank. The gas flow rate to the frit was monitored during the course of all flotation experiments. A launder around the top of the cell enabled rapid removal of the froth to maximize froth efficiency.

For flotation experiments, 100 g of hydrophobic quartz particles were floated in 10^{-1} M KNO_3 at pH 5.6 over 8 min. Concentrates were collected at 0.5, 1.5, 4, and 8 min (cumulative). The flotation rate constant (k) as a function of particle size was determined by fitting a first-order rate equation ($R = R_{\max}(1 - e^{-k't})$), using a nonlinear least-squares regression, to the recovery versus time curves obtained for each particle size. The flotation rate constant at time zero (k) is the product of the flotation rate constant k' and the maximum flotation recovery, $R_{\max}$, derived from the fit of the experimental data with the first-order rate equation. Particle size measurements were conducted using a Malvern Master-sizer MSX14. In these experiments, great care was taken to ensure that the pulp was properly mixed. Only above specific impeller speeds and gas flow rates did this occur, verified by measuring pulp densities and particle size distributions at various depths in the cell. When both sets of data were constant with depth, the criterion of efficient mixing was satisfied. All data are reported only under these circumstances.

3.4. Bubble size determinations

A University of Cape Town (UCT) bubble size analyzer was used to determine the bubble size in the flotation cell. The UCT bubble size analyzer captures the bubbles and draws them through a glass capillary (diameter 0.4 mm). Diodes measure the change in refractive index between the water and gas. Using these readings and the volume of gas collected in a burette, the bubble size distribution can be obtained. Up to 4000 bubbles were typically measured using the analyzer over a period of 90 s, permitting a statistically significant number of bubbles to be counted. Measurements were performed in duplicate and the analyzer data were then compared with photographs. Agreement between the two techniques was very satisfactory, with an overall reproducibility of $\pm 0.05 \text{ mm}$ in bubble diameter.

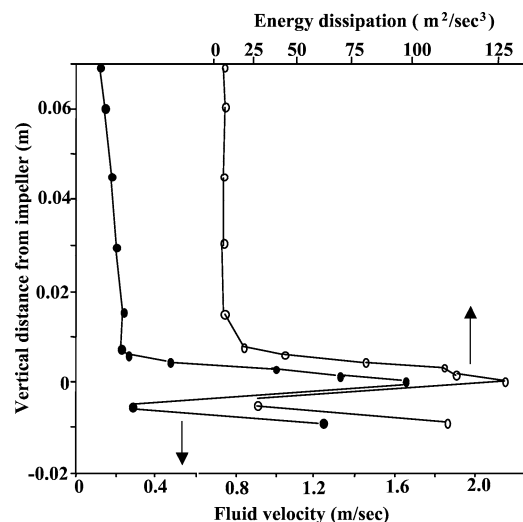


Fig. 2. Fluid velocity, V_f (filled circles), and turbulent dissipation energy, ε (empty circles), as functions of vertical distance from the impeller in the Rushton flotation cell and at a fixed distance of 0.026 m away from the impeller shaft (agitation 650 rpm; $G_{fr} = 3.5 \times 10^{-3} \text{ m}^3 \text{ min}^{-1}$).

3.5. Laser Doppler velocimetry

Laser Doppler velocimetry (LDV) measurements were carried out in the Rushton turbine flotation cell to determine the fluid velocity in the cell under the same conditions (gas flow rate and agitation speed) as in the flotation experiments with quartz. Measurements of velocity and velocity fluctuations were carried out at different heights and depths into the cell, and angles of rotation in one of the cell quadrants. Measurements were conducted in the Rushton turbine flotation cell at varying heights with the measurement position weighted towards the impeller, as this region is of great importance, as observed in Fig. 2 and discussed by Rutherford et al. [53] and Wu and Patterson [54]. Rutherford et al. [53] showed that three stable flow patterns resulted (parallel flow, merging flow, and diverging flow), depending upon the distance between the bottom impeller and the floor, the top impeller and the pulp/air interface, and the two impellers, in the Rushton turbine flotation cell. For example, parallel flow results from the impeller being placed one-third of the height of the cell above the cell floor [55].

From the LDV measurements, the average fluid velocity, V_f , and the turbulent dissipation energy, ε , were determined. The value of V_f is determined from velocity measurements in the x , y , z directions:

$$V_f = \sqrt{V_x^2 + V_y^2 + V_z^2}. \quad (36)$$

It was assumed in this study that the velocities of bubbles (v_b) and fluid (V_f) are similar. The turbulent kinetic energy, q , is determined from the root mean square of the fluctuations of the turbulent fluid velocities $\bar{U}_x$, $\bar{U}_y$, and $\bar{U}_z$,

$$q = \frac{1}{2} [\bar{U}_x^2 + \bar{U}_y^2 + \bar{U}_z^2]. \quad (37)$$

In turbulent conditions [51] the turbulent dissipation energy can then be calculated from

$$\varepsilon = \frac{0.85q^{3/2}}{L}, \quad (38)$$

where L is a measure of the turbulent length macroscale, generally assumed constant and equal to half the impeller blade width (in this study $L = 0.005$ m).

4. Results and discussion

Figure 2 shows an example of the LDV measurements of fluid velocity, V_{fl} and turbulent dissipation energy, ε , measured at a fixed distance of 0.026 m away from the impeller shaft and at various heights inside the Rushton flotation cell (agitation 650 rpm; $G_{fr} = 3.5 \times 10^{-3} \text{ m}^3 \text{ min}^{-1}$). As expected, the values of V_{fl} and ε are higher close to the impeller, independently of agitation speed and gas flow rate. They remain more or less constant from a height of 0.01 m above the impeller. All the values measured in this study in the Rushton flotation cell are also in agreement with those reported in the literature for similar types of flotation cells [53,54].

Figure 3 shows some results of calculation for E_c , E_a , E_s and flotation rate constant, k , using Eqs. (20), (22), (34), and (15), respectively, as a function of quartz particle diameter and induction time (Eq. (26)). For the induction time we have used the values of the parameters A and B in Eq. (26) obtained in our previous study of the attachment efficiency of quartz particles [35]. In this latter study, it was found that the value of B was 0.6 ± 0.1 and independent of particle contact angle, while A had values of 0.051, 0.058, and 0.074 for methylated quartz particles with advancing water contact angle of 80° , 65° , and 50° , respectively. E_c increases

with particle size, while both E_a and E_s decrease with increasing particle size but also with increasing induction time. The effects of quartz particle size and hydrophobicity (induction time) on the experimental and calculated collision and attachment efficiencies have already been discussed in previous studies [3,4,26,35]. The combination of these efficiencies in Eq. (15) produces the characteristic shape of the flotation rate constant versus particle size curve with a maximum value obtained at intermediate particle size values. A maximum flotation rate constant value of 5 min^{-1} is typical for fully liberated and hydrophobic sulfide particles in a size range between 50 and $70 \mu\text{m}$ [6]. The low values of flotation rate constants obtained on each side of this maximum are of course the results of low collision efficiencies for fine particles, and low attachment and stability efficiencies for coarse particles.

Figure 4 shows the effect of dissipation energy on these efficiencies and flotation rate constant. With increasing dissipation energy, E_s decreases as a result of a larger acceleration force on the particle–bubble aggregate while the flotation rate constant increases because of increased particle–bubble collision frequency. Figure 5 shows the effect of bubble velocities on these efficiencies and flotation rate constant. Increasing bubble velocity has a small effect on particle–bubble collision efficiency but a large effect on attachment efficiency and collision frequency, especially for the large particles. A comparison of the results in Figs. 3–5 indicates that the flotation rate constant of large particles is very sensitive to changes in surface hydrophobicity and hydrodynamic conditions inside the flotation cell, and more particularly to changes in bubble velocity.

The adequacy of these models in describing the flotation of quartz particles in a Rushton flotation cell is tested in Fig. 6 for the case of particles with an advancing water contact angle of 80° (parameters in Eq. (26): $B = 0.6$ and $A =$

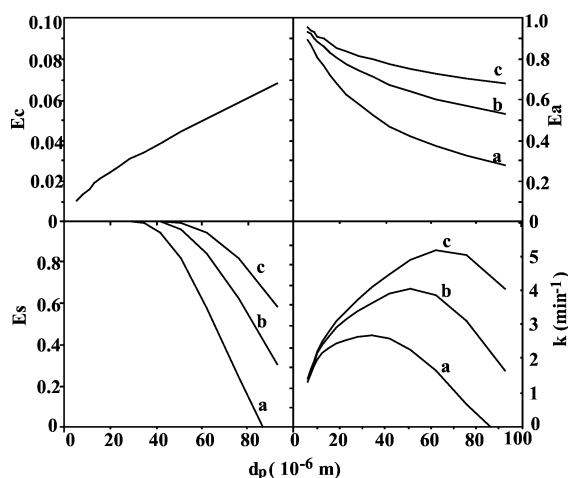


Fig. 3. Calculated collision (E_c), attachment (E_a), and stability (E_s) efficiencies and flotation rate constant (k) as a function of quartz particle diameter and induction time parameters in Eq. (26): $B = 0.6$ and $A =$ (a) 0.074, (b) 0.058, and (c) 0.051 ($d_b = 1.4 \times 10^{-3} \text{ m}$; $v_b = 0.18 \text{ m s}^{-1}$; $G_{fr} = 3.5 \times 10^{-3} \text{ m}^3 \text{ min}^{-1}$; dissipation energy $38 \text{ m}^2 \text{ s}^{-3}$).

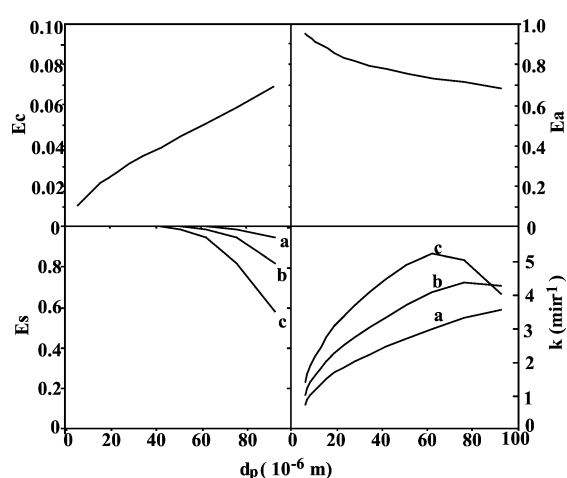


Fig. 4. Calculated collision (E_c), attachment (E_a), and stability (E_s) efficiencies and flotation rate constant (k) as functions of quartz particle diameter and dissipation energy of (a) 10, (b) 20, and (c) $38 \text{ m}^2 \text{ s}^{-3}$ (advancing water contact angle = 80° ($A = 0.051$ and $B = 0.6$ in Eq. (26))); $d_b = 1.4 \times 10^{-3} \text{ m}$; $G_{fr} = 3.5 \times 10^{-3} \text{ m}^3 \text{ min}^{-1}$; $v_b = 0.18 \text{ m s}^{-1}$).

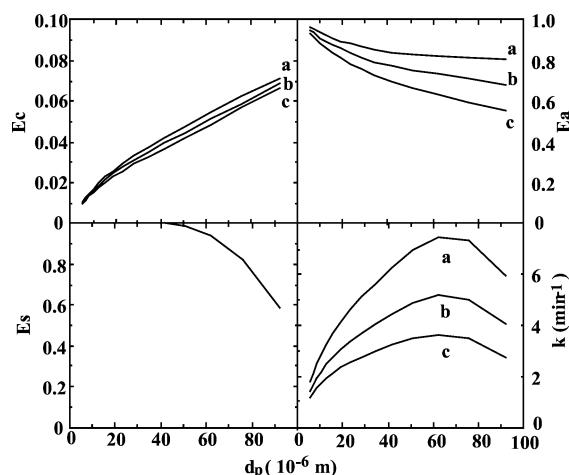


Fig. 5. Calculated collision (E_c), attachment (E_a), and stability (E_s) efficiencies and flotation rate constant (k) as functions of quartz particle diameter and bubble velocity of (a) 0.15, (b) 0.18, and (c) 0.21 m s^{-1} (advancing water contact angle = 80° ($A = 0.051$ and $B = 0.6$ in Eq. (26)); $d_b = 1.4 \times 10^{-3}$ m; $G_{fr} = 3.5 \times 10^{-3}$ $\text{m}^3 \text{min}^{-1}$; dissipation energy $38 \text{ m}^2 \text{s}^{-3}$).

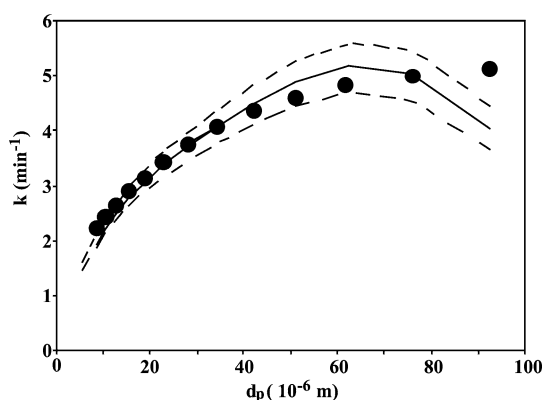


Fig. 6. Comparison between the calculated (lines) and experimental (circles) flotation rate constants as a function of quartz particle diameter. The broken lines correspond to the 95% confidence limits of flotation rate constants calculated with values of the two variable parameters: $v_b = 0.18 \pm 0.01 \text{ m s}^{-1}$ and dissipation energy = $38 \pm 7 \text{ m}^2 \text{s}^{-3}$. (Experimental parameters: $d_b = 1.4 \times 10^{-3}$ m; agitation 650 rpm; $G_{fr} = 3.5 \times 10^{-3} \text{ m}^3 \text{min}^{-1}$; advancing water contact angle 80° ; $A = 0.051$ and $B = 0.6$ (Eq. (26)).)

0.051) interacting in a Rushton flotation cell with gas bubbles 1.4×10^{-3} m in diameter introduced at a gas flow rate of $3.5 \times 10^{-3} \text{ m}^3 \text{min}^{-1}$ and at an agitation speed of 650 rpm. The best fit of the experimental flotation data with Eq. (15) is obtained with a fluid velocity of $0.18 \pm 0.01 \text{ m s}^{-1}$ and a dissipation energy of $38 \pm 7 \text{ m}^2 \text{s}^{-3}$ (95% confidence limits). These values are relatively low and correspond to values measured in a region inside the flotation cell away from the impeller (cf. Fig. 2). This observation is consistent with Schulze's earlier prescient suggestion that the mean vessel energy dissipation determines the mean stability of the aggregates, rather than the local dissipation energy. We conclude that the conditions of high turbulence around the impeller are conducive to low particle–bubble stabilities.

5. Conclusions

An analytical model that enables the calculation of the flotation rate constant of particles as a function of particle size with, as input parameters, measurable particle, bubble, and hydrodynamic quantities has been derived. This model is similar to that for nonturbulent flow conditions, except for the inclusion of the turbulent dissipation energy which affects both the stability efficiency and the frequency of collisions between particles and bubbles. In this expression, the generalized Sutherland equation collision model and the modified Dobby–Finch attachment model were used to calculate the efficiencies of particle–bubble collision and attachment, respectively. Both models have been shown previously to adequately describe the collision and attachment of methylated quartz particles with gas bubbles in potential flow conditions over a large range of particle size and contact angle, bubble size and electrolyte concentration. The stability efficiency model includes the various forces acting between the bubble and attached particle and depends mainly on the relative magnitude of particle contact angle and turbulent dissipation energy. Its contribution affects the flotation of coarse particles in the main.

At low turbulent dissipation energies, the flotation rate constant of particles with moderate to high contact angles can be essentially described by the collision efficiency model. The curves of flotation rate constant versus particle size increase almost linearly with particle size. On the other hand, with high turbulent dissipation energies, the flotation rate constants increase in magnitude as a result of increased bubble–particle collision frequencies, but also decrease due to bubble–particle instability. Since the latter effect is of greater magnitude for the coarser particles, these calculations produce the characteristic shape of the flotation rate constant versus particle size curve, with a maximum obtained at intermediate particle size, seen in practice.

The flotation rate constants calculated with these models were compared to the experimental flotation rate constants of methylated quartz particles with particle diameters between 8 and 80 μm with an advancing water contact of 80° , interacting in a Rushton flotation cell with gas bubbles 1.4×10^{-3} m in diameter, introduced at a gas flow rate of $3.5 \times 10^{-3} \text{ m}^3 \text{min}^{-1}$ and at an agitation speed of 650 rpm. Good agreement between the calculated and experimental flotation rate constants was obtained when the values of bubble velocity and turbulent dissipation energy used in the calculations corresponded to those measured in a region of low turbulence inside the flotation cell away from the impeller.

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Appendix A. Nomenclature

a	centrifugal acceleration of particle–bubble aggregate ($= a_{\text{cen}}$)
d_b	bubble diameter
d_p	particle diameter
d_{pb}	sum of particle and bubble radii
E_a	attachment efficiency
E_c	collision efficiency
E_{coll}	collection efficiency
$E_{c\text{-su}}$	Sutherland collision efficiency
E_{GSE}	generalized Sutherland equation collision efficiency
E_s	stability efficiency
F_b	buoyancy force
F_{ca}	capillary force
F_d	acceleration force
F_g	gravitational force
F_{hyd}	hydrostatic force
F_{σ}	capillary pressure force
$\bar{G}$	mean velocity gradient in a turbulent fluid
G_{fr}	gas flowrate
h	flotation cell height
J_g	superficial gas velocity
N_p	number of particles per unit volume
N_b	number of bubbles per unit volume
q	turbulent kinetic energy
r_b	bubble radius
Re_b	Reynolds number
r_p	particle radius
S_b	bubble superficial surface area rate
t_{ind}	induction time (calculated from Eq. (26) with parameters A and B)
t_r	residence time of bubbles in unit volume
t_{sl}	sliding time of particle on bubble
V	unit volume
v_b	bubble velocity
$\bar{V}_b$	velocity of bubbles relative to fluid velocity
V_{cell}	flotation cell volume
V_{fl}	fluid velocity
v_p	particle velocity
$\bar{V}_p$	velocity of particles relative to fluid velocity
Z_{pb}	collision frequency between particles and bubbles
γ	liquid–vapor surface tension
ε	turbulent dissipation energy
θ	collision angle of particle on bubble
θ_a	adhesion angle
θ_t	maximum possible collision angle of particle on bubble (angle of tangency)
ρ_b	bubble density
ρ_{fl}	fluid density
ρ_p	particle density

ν	kinematic viscosity
φ	contact angle

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